# Brahin Uddin

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## Education

**Rutgers University–New Brunswick** Expected May 2028
*B.S. Electrical Engineering, Minor in Computer Science* *New Brunswick, NJ*

- GPA: 3.6 – Dean's List, Honors College, Engineering Honors Academy
- Relevant Coursework: Computer Architecture, Digital Logic Design, Semiconductors, Electronic Devices, Linear Systems & Signals, PCB Design, Data Structures, Discrete Structures

## Projects

**RV32IM Five-Stage Pipelined RISC-V Core** Mar 2026 – Jul 2026
*SystemVerilog, Verilator, cocotb, Spike, Vivado – Xilinx Zynq-7020 SoC* *Personal Project*

- Designed an in-order RV32IM + Zicsr core (IF/ID/EX/MEM/WB) in SystemVerilog with full operand forwarding, a load-use hazard interlock, precise exceptions, and a DSP48-based pipelined multiplier
- Verified against the official riscv-tests and RISCOF compliance suites (48/48 passing) and instruction-by-instruction lockstep comparison with the Spike reference model over 10.0M instructions, zero divergences
- Ran on Zynq FPGA fabric with the ARM as loader and console, scoring 2.15 CoreMark/MHz at 150 MHz (CPI 1.18), with hardware cycle counts matching simulation exactly

**Low-Latency Ethernet Protocol Decoder** Jul 2026 – Present
*SystemVerilog, cocotb, Vivado, Python – Xilinx Zynq-7020 SoC* *Personal Project*

- Built a Nasdaq TotalView-ITCH 5.0 parser in SystemVerilog decoding Ethernet/IPv4/UDP/MoldUDP64 in FPGA fabric, reaching 73.5 ns median latency at 3.39M msgs/sec (line rate) in simulation via cut-through parsing with FCS rollback
- Designed a Gray-coded async FIFO for clock-domain crossing (125 MHz PHY to 200 MHz core); verified the design against a Python golden model in a cocotb testbench with fault injection, 7,430 records, zero mismatches
- Synthesized and closed timing in Vivado at 200 MHz (WNS +0.412 ns, WHS +0.041 ns) using 2,557 LUTs and 2,509 FFs (4.8% of device), with RGMII input capture in IDELAYE2/IDDR I/O primitives

## Experience

**Data Center Facility Operations Intern** May 2026 – Aug 2026
*Crusoe* *San Francisco, CA*

- Built a Python/JavaScript pipeline that parsed raw BMS/EPMS telemetry JSON into clean, structured datasets, cutting on-site processing time from hours to minutes
- Evaluated ML and reinforcement-learning approaches for the predictive control layer of a closed-loop digital twin of a remote data center, and modeled cost tradeoffs between physics-only and hybrid physics+ML training pipelines
- Built the CMMS asset hierarchy for 700+ electrical and mechanical components and authored RFPs and MEP equipment lists, translating engineering drawings into buildable specifications

**Low Voltage Systems Engineer** Sep 2024 – Present
*Rutgers Solar Car* *Piscataway, NJ*

- Engineered a daisy-chain FDCAN communication topology linking 8 distributed system boards, enabling reliable real-time data exchange between high/low-voltage managers and peripheral modules
- Designed the "LV3 Core" mixed-signal carrier PCB (STM32G4 MCU) for low-voltage battery telemetry and protection
- Implemented reverse-polarity protection, voltage-sensing telemetry, and buck-converter regulation, then validated signal integrity and I/O pin mapping during hardware bring-up

**Software/Controls Engineer** Sep 2024 – Present
*Space Technology Association of Rutgers (STAR)* *Piscataway, NJ*

- Designed a custom PCB interfacing an NVIDIA Jetson Nano onboard computer with peripheral sensors and motor controllers, engineering signal-routing and power-distribution circuitry for reliable multi-sensor operation
- Developed C/Python control logic and motor-driver controls for a NASA Lunabotics lunar mining robot on Linux
- Implemented SLAM-based 3D perception and autonomous navigation using a Unitree 4D Lidar L2 and ROS 2

## Technical Skills

- **FPGA & RTL:** Digital Design, RTL, Functional Verification, Synthesis, Timing Closure, CDC; Verilog, SystemVerilog, Vivado, ILA Debug, cocotb, Verilator, Yosys
- **Interfaces & Protocols:** Ethernet MAC (RGMII/GMII, FCS/CRC-32), UDP/IPv4, MoldUDP64, AXI4-Lite, FDCAN
- **Programming & Tools:** C, Python, Tcl, RISC-V Assembly, MATLAB, Java, Linux, Git/GitHub, SQL, ROS 2
- **PCB & CAD:** KiCad, LTSpice, Mixed-Signal Design, SolidWorks/OnShape, Oscilloscopes, Soldering, 3D Printing